GRISHMA BALGI

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Applied ML & Sensor Intelligence

EDUCATION

New York University | May 2027
M.S. Mechatronics and Robotics | GPA: 3.834/4.0

Veermata Jijabai Technological Institute | June 2025
B.Tech. Mechanical Engineering; Honors in Automobile Engineering | GPA: 9.04/10

TECHNICAL SKILLS

Programming & Data: Python, NumPy, Pandas, Matplotlib, scikit-learn

Modelling: Classification, feature preprocessing, model evaluation, reduced-order models

Tools & Hardware: Altair romAI, PyTorch/TensorFlow exposure, Arduino, sensor acquisition

EXPERIENCE

Prema Lab, New York University | Sep. 2025 - Present
Graduate Research Assistant | Robotics fabrication, instrumentation and simulation

- Fabricated a robotic-fin test rig and automated Silhouette Portrait 4 cutting through Python USB commands, generating coordinate-based paths for candidate fin geometries.
- Developed a motor-control interface and Arduino acquisition firmware to vary oscillation waveform, amplitude and frequency; logged load-cell readings alongside commanded and measured motor positions.

Schneider Electric | June 2024 - June 2025
Engineering Intern | Mechanical modelling and sensor systems

- Explored data-driven reduced-order models using Altair romAI, preparing simulation outputs and evaluating model predictions for mechanical systems.
- Investigated vibration sensing and Python-based condition-monitoring workflows for circuit breakers, connecting physical measurements with model development.

SELECTED PROJECTS

Multi-Sensor Circuit-Breaker Fault Detection
Python, Arduino, scikit-learn, sensor acquisition | Undergraduate team project | 2024-2025

- Developed a circuit-breaker monitoring prototype using vibration and acoustic sensing, sensor-data preprocessing and classification of spring-removal conditions.
- Compared classical classifiers and developed a CSV-based prediction workflow; kept synthetic-data experiments distinct from physical measurements and deployment claims.

Camera-Only Visual Navigation
Python, RootSIFT, VLAD, graph search | Course project | In progress

- Building a camera-only navigation pipeline using visual descriptors, a topological graph and Dijkstra search to navigate toward image-defined goals.
- Working on place recognition and geometric verification; evaluating extensions separately from the baseline rather than treating proposed methods as completed results.